

UCL SGNL OPS / Orbit Dynamics Library

Assessed from the working tree at ~/odl20lite, x86-64 Linux, GCC 15

Verdict

The ray-traced radiation pressure assets are conclusively gone — and in the same week the dynamics were validated at **0.33 m against IGS final orbits, out-of-sample**, with the satellite's area-to-mass ratio recovered independently from the fit. That combination moves the recommendation off “archive” and onto one narrow, evidenced play. There is no propagator to sell. There is a measured claim to own about non-conservative force modelling, and the next number that matters is what an estimated radiation pressure parameter recovers.

What changed since revision 1

Revision 1 rested on two open questions. Both are now answered, and they pull in opposite directions.

The crown jewel is gone, conclusively. A search of all of /home, both Windows drives, the Dropbox tree and .attic/ found zero .grd files, none of the eight mission grid-file directories, no ray-tracer, and no SRP_TRR_5_0_7. The Dropbox copy is the origin of this working tree and is identical in this respect, so the assets were never in the synced tree rather than pruned from a working copy.

The dynamics are validated, and the validation is good. Fitting to IGS final precise ephemerides now gives 0.334 m position RMS on a 21.75 h GPS arc, out-of-sample. Revision 1 treated the commodity propagation layer as dead weight. That was wrong, and the correction is the single most important change in this revision.

What I measured

Validated accuracy	0.334 m position RMS, GPS G01, 21.75 h arc, out-of-sample; 0.343 m in-sample. IGS finals are good to ~ 2.5 cm, so they are truth here
Recovered physics	An area sweep finds a clear minimum at $A/m = 0.0209 \text{ m}^2/\text{kg}$, close to the published GPS figure — recovered from the fit, not assumed
Integrator quality	RK8/7 against the analytic two-body solution: 1.48 cm over one sidereal day at GEO, and step-independent at 60 s / 10 s / 1 s, so a formulation offset rather than truncation error
Throughput	0.81 s per one-day arc at 70×70 gravity + third body + solid and pole tide + GR, 10 s step
Build state	12 of 12 smoke checks pass, x86-64 Linux / GCC 15
Size	15,107 LOC library (54 files), 4,764 LOC headers, 13,863 LOC vendored
Authors	David Harrison (101 files), Santosh Bhattarai (40), Zhen Li, Stuart Grey, Alex Forsyth, Marek Ziebart — 2014–2017

The validation result, and why it is now the asset

Turning single force models off on the G01 arc shows where the error lives:

change	position RMS
full model	0.34 m
spacecraft area left at default	22.14 m
solar radiation pressure off	31.51 m
third-body gravity off	356.80 m
gravity degree 4 instead of 12	indistinguishable
gravity degree 20 instead of 12	indistinguishable

Three things follow, and they set the whole strategy.

Luni-solar attraction dominates at MEO and is modelled correctly — removing it costs 357 m. **The gravity field is effectively a point mass up there**, so there is nothing to gain from a higher expansion for GNSS work. **Everything else is swamped by solar radiation pressure**, which is precisely where the scarce capability was supposed to live.

And the confirming case: a second GPS satellite, G05, fits to 3.200 m rather than 0.334 m, because it is a different block and the hand-set area does not suit it. The error there is not the propagator. It is the quality of the spacecraft *description*.

That is what makes the validation commercially load-bearing rather than merely reassuring. It is not a propagator anyone would buy — Orekit is free and better governed. It is the evidence base that makes a claim about radiation pressure modelling falsifiable. Nobody takes a radiation pressure model seriously when it is bolted to dynamics of unknown quality, and until this week the dynamics here were of unknown quality.

The crown jewel: absent, and how absent

Eight grid-file directories are named in `src/Resident_constants.cpp`. None exist. No ray-tracer is present; nothing in the tree *writes* a `.grd`.

Two qualifications matter, one in each direction.

Softer than “capability lost”. The grid file format is open — Golden Software Surfer 6 ASCII — and the reader in `Force_rp_gridfile.cpp` is intact and working. `scripts/surfer6poleReplicate.py` survives and documents the grid convention in full (361×181 nodes over $[-180, 180] \times [-90, 90]$). The 27 curated per-satellite optical and thermal property sets that *parameterise* a ray-trace are still in source. The pipeline shape is known: ray-trace at spiral sample points, pad, grid, replicate poles.

Harder than “a bounded engineering job”. The file format was never the difficult part. The difficult parts are validated spacecraft CAD geometry with per-surface materials, and the ray-tracing physics over it — self-shadowing, multiple reflection, specular and diffuse separation per surface. None of that is in the tree, and GNSS spacecraft geometry is largely not public. The gap is bounded in specification and unbounded in content. Writing a Surfer 6 emitter is the easy tenth of the job.

What has been closed since revision 1

Four of the blockers in revision 1 are now gone. This matters because it is what makes the asset transactable at all.

- **Version control.** The tree is under git, with the 559 MB of atmosphere tables, a 391 MB Visual Studio cache and 1,850 Dropbox sidecar files excluded and documented rather than committed.
- **Provenance.** `PROVENANCE.md` inventories six vendored components and ten third-party

datasets with their licences and obligations.

- **A silent correctness defect.** The orbit fitter was building least squares on identity partials, because only two of the five propagators integrate the state transition matrix and the default is not one of them. It now refuses rather than converging quietly on a wrong answer.
- **A latent build defect.** The Makefile tracked no header dependencies, so editing a header recompiled only the changed translation unit and left every other object with the old struct layout — corrupting memory at run time rather than failing to compile. Any measurement taken before this fix was untrustworthy.

What still blocks a transaction

FOR THE USER not solvable in code

- **No licence or copyright statement anywhere in the first-party code**, and the work originates from UCL SGNL. Under standard UK university IP terms it belongs to UCL, not to whoever holds the tree. Nothing can be sold, licensed or released until this is settled.
- **CALCEPH is triple-licensed** — CeCILL-B (BSD-like), CeCILL-C (LGPL-like) or CeCILL v2.1 (GPL-like) — a licensee must choose one, and no choice is recorded. The Makefile links it *statically* into every binary, so under two of the three that choice propagates. It is optional and the build drops it cleanly, so shipping without it is a real alternative. This is a decision for UCL or for you, not for an engineer.
- FECsoft and SGP4 are book companion code carrying no licence text in this tree.

ENGINEERING ranked by what they cost

- **Estimation is six-state only.** No radiation pressure scale factor, drag coefficient or empirical accelerations — exactly what operational orbit determination estimates, and exactly what the G05 result says is the binding constraint. *This is now the highest-value open item and is commissioned.*
- **No observation models.** No pseudorange, carrier phase, radar or optical angles. It fits to already-determined precise orbits. It is a force-model validation tool, not an orbit determination system.
- **No covariance.** Rules out conjunction assessment and space situational awareness.
- No manoeuvre modelling. Single-threaded, one object per process. No Python bindings. IAU1980 frames rather than IERS 2010. Regression baselines still to be stored.

Valuation

- **The library as a sellable product: still approximately £0.** Nothing in this revision changes that. Orekit, GMAT, GODOT and Tudat are free, better governed and better tested, and they are not the competition to beat — they are the reason not to compete here at all.
- **The validation as a credential: materially positive, and new.** A reproducible 0.33 m out-of-sample figure with the physics recovered independently is the difference between an unfalsifiable claim and a defensible one. It cost nothing to obtain because the dynamics were already correct; it was simply never measured.
- **Know-how and services: still the only line item that could carry a business**, but now with evidence behind it rather than reputation alone. UCL-lineage non-conservative force modelling sold as analysis to GNSS operators, altimetry missions, or SSA firms needing better radiation pressure and area-to-mass handling for high area-to-mass ratio debris.
- **Replacement cost remains the wrong frame.** Roughly £300k–£600k to rebuild, but you would not rebuild it — you would adopt Orekit and port the differentiated physics onto it.

Recommendation

Neither archive it nor productise it. Pursue one narrow, evidenced capability play, and let the next measurement decide whether it goes further.

1. **Estimate radiation pressure as a parameter** rather than hard-coding an area per satellite. This is commissioned and in progress. It converts a six-state curve fit into an orbit determination tool, should bring G05 into line with G01, and — the reason it comes first — *it measures how much of the residual error a ray-traced model could ever recover*. Until that number exists, nobody can say what a rebuilt generator is worth.
2. **Broaden the validation table** across GPS blocks, Galileo, and a LEO arc. Each row is cheap now and the table is the asset a customer or a funder actually reads.
3. **Hold the ray-tracer.** Deliberately not commissioned. The binding constraint is spacecraft geometry that is absent and largely non-public, and the prize is unknown until step 1 reports. Scope it on paper; build it only if the residual after step 1 justifies it.
4. **Settle the IP position in parallel.** It gates everything else and it is the one workstream that needs no engineering at all.

What to find out first

Revision 1 asked three questions. One is answered — the ray-traced assets and their generator are gone. Two remain, and both are yours rather than an engineer's.

1. **Who owns this?** UCL's position on the 2014–2017 code, and whether anything can be licensed or released. Also the CALCEPH licence choice, which is a smaller instance of the same question.
2. **Is there a named customer for GNSS-grade radiation pressure and thermal re-radiation analysis?** If yes, the validation table is the opening argument and the asset is a services business. If no, it is a well-documented archive with an unusually good accuracy figure attached.

Basis: measured facts come from the tree at `/home/rog/odl20lite` and from runs executed on 15 September 2026, including the Phase 1 investigation carried out by the `odl_maintainer` session. The competitive and valuation sections are judgement, not repository evidence. Revision 2 supersedes revision 1, which understated the commodity layer.